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RapidAlignerSeismic: A Python package for seismological data mining accelerated on GPUs

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Summary

RapidAlignerSeismic (RAS) is a Python package that uses GPUs to accelerate template matching procedure on seismic data. It is designed from NVIDIA rapidAligner (Hundt et al., 2014). The package is designed for processing long streams of three-component continuous seismic data stored in a database (structured or not) in the miniseed format with a minimum amount of data preprocessing and user input. The I/O procedures use the Python ObsPy package (Beyreuther et al., 2010). Template matching involves selecting a set of earthquake waveforms (templates) that are correlated with the continuous data streams in order to detect similar events. Compared to catalogues built by human analysts, the resulting enhanced catalogues are typically an order of magnitude larger. RAS also provides a set of tools for post-treatment designed to identify and visualize clusters of seismic events with similar waveforms. RAS can also be used in a so-called Systematic mode, which automatically detects any repeating patterns within continuous data streams. This allows users to obtain automatic catalogues of earthquakes without prior knowledge of the seismicity in the region of interest. RAS is targeted toward seismologists working with seismic data.

Statement of Need and State of the Field

In recent years, the field of seismology has undergone a revolution, driven by a rapid increase in the volume of seismic data. Big data seismology (Arrowsmith et al., 2022) has therefore become an important emerging field which aims to develop innovative approaches to efficiently extract valuable information from these vast datasets. While earthquake detection remains an important task in modern seismology, researchers are increasingly interested in other types of signals in seismic data, such as long-period events (LPE), volcanic tremors (VT) and non-volcanic tremors (NVT), which offer invaluable information spanning a wide spectrum of geophysical phenomena.

To improve and automate earthquake detections, an effective strategy involves cross-correlating a short waveform template with a continuous data stream (Anstey, 1964; Beaucé et al., 2018; Frank et al., 2017; Gibbons & Ringdal, 2006; Senobari et al., 2019; Shelly et al., 2007). Such a template is typically a three-component waveform of a previously detected and well-localized earthquake. This template matching technique yields a continuous correlation function, quantifying the similarity between the template and the data stream. By combining information from adjacent stations, the detection threshold can be considerably reduced, allowing detection of very small earthquakes, often missed by human analysts (e.g., Li & Zhan (2018)). Such events, called repeaters, share the same focal mechanism and occur within a

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short distance from the reference template event (e.g.,Waldhauser & Ellsworth (2000); Schaff & Waldhauser (2005)).

The template matching method requires a large number of operations as it involves correlating three-component signals sampled at 100 Hz over periods ranging from one to several years with 20-second templates. For reference, processing one day requires computing about 26 million Coss Correlation (CC) scores, and this figure scales with the number of stations, days, and templates to be processed. Hence, there is a need to develop fast, parallel algorithms that leverage GPU capabilities, such as RapidAlignerSeismic, to process years of seismic data in a reasonnable timeframe.

The enhanced efficiency of RAS enables the direct processing of high-frequency, raw seismic data, extracted directly from seismic databases for automatic template detection. In this Systematic mode (see bellow for details), each time window of a long continuous data stream is correlated with all the others in order to detect matching pairs of similar signals. This blind and massive analysis of continuous seismic data provides enhanced catalogs of events, which can then be used to detect similar events over extended periods of time using a classical template matching approach.

Since RAS analysis can generate thousands of detections, it is impossible to examine them all manually. Therefore, we have developed clustering tools that allow for the classification of detections, the creation or enrichment of an event catalogue, and easy visualization of detections per cluster.

Over the past years, several tools have been developped, such as the Fast Matched Filter (FMF) (Beaucé et al., 2018), EQcorrscan (Chamberlain et al., 2017), and the Fingerprint And Similarity Thresholding (FAST) (Yoon et al., 2015). The key features that differentiate RapidAlignerSeismic from those template matching tools are the following:

- RAS is a complete, user-friendly, end-to-end package that performs complete template matching and post-treatment analysis.
- RAS is designed to process seismic data directly from a seismic database without any format change or subsampling of the data.
- Two computing modes: Standard and Systematic. The Systematic mode requires no prior templates, enabling the extraction of any type of seismic signals from the data and the creation of seismicity catalogues frow scratch.
- We use a more robust method to detect outliers based on the mean absolute deviation ($\widehat{MAD}$) estimator.
- RAS provides tools to analyze the results of template matching and detections classification by clustering.

The RapidAlignerSeismic package

The RAS package provides a comprehensive set of tools for template matching analyses. The online documentation provides details on how to setup and perform a template matching analysis, as well as analyzing the results.

RapidAlignerSeismic can be run in two modes: Standard or Systematic mode (cf. Figure 1). In Standard mode, a classical template matching is performed using a predefined catalogue of events, or on a catalogue of templates obtained in Systematic mode. Systematic mode involves a blind search for repeating signals in a long (typically several months), continuous three-component data stream. This mode is particularly useful when there are no pre-existing template databases or when data mining is used to search for new types of seismic signals.

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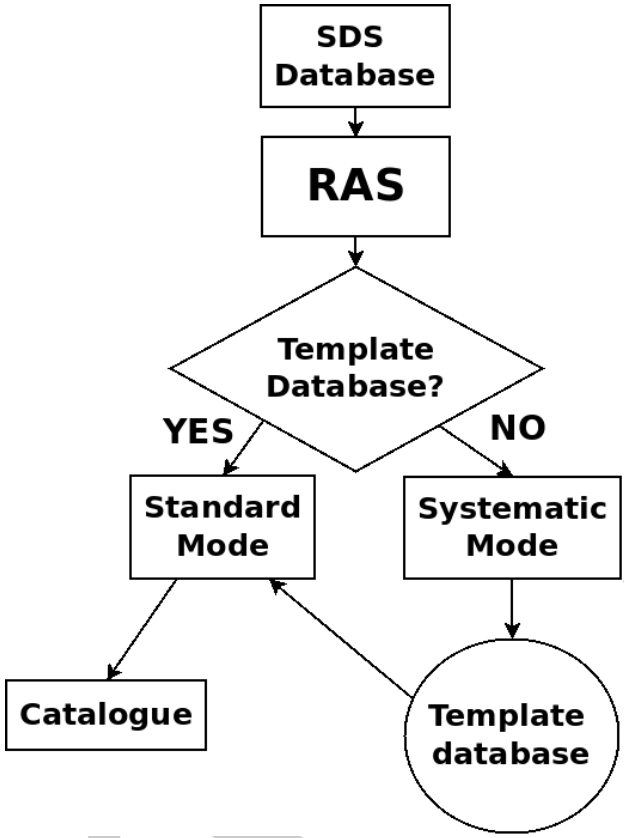


Figure 1: Schematic view of the workflow of RapidAlignerSeismic.

In Standard mode, RAS computes the normalized correlation coefficient (CC) vector (see (Hundt et al., 2014) for details) for a given template and three-component daily miniseed files. To identify a seismic event in the CC vector, RAS uses a thresholding method based on the Median Absolute Deviation ($\widehat{MAD}$)(Efron & Tibshirani, 1994; Rousseeuw & Hubert, 2011). First, RAS computes the Fisher transform of the stacked CC vector (i.e. the normalized sum of the CC vector for each component of the seismic signal):

$$z = 0.5 * log((1 + CC)/(1 - CC))$$

The $\widehat{MAD}$ is then defined by:

$$\widehat{MAD} = 1.483 * median(|z - median(z)|)$$

The detection threshold is now defined as a multiple m of the $\widehat{MAD}$, where m is defined by the user in the input file. A typical choice for m ranges from 8 to 10 (Muir et al., 2023).

In Systematic mode, RAS first subdivides the daily data files into smaller segments typically ranging from 20 to 30 seconds in length, as defined by the user in the input file. Then, RAS computes the CC vector for each segment relative to the rest of the signal. Figure 2 illustrates this process by showing the extraction of a 20-second segment from the daily time series and the computation of the corresponding CC vector against the rest of the signal for that day. Figure 3 shows the corresponding CC vector, which starts after the extracted event and is computed until the end of the day. After computing the CC vector for a day, RAS checks for detections using a fixed threshold. For a given segment, if multiple events are detected, only

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the best one is kept. If no matching event is detected, RAS compares the segment to the next day, and continues until a matching event is detected or the end of the provided data is reached. Once a segment is processed, RAS moves on to the next one and so on. This method reduces the computing time and limits the number of duplicate events to be removed during post-processing.

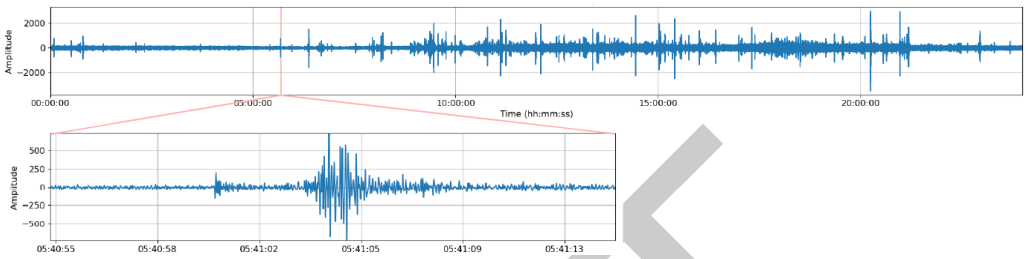


Figure 2: Example of a 20 s-long signal (bottom panel) extracted within a 24 h-long data stream (top panel), sampled at 100 Hz

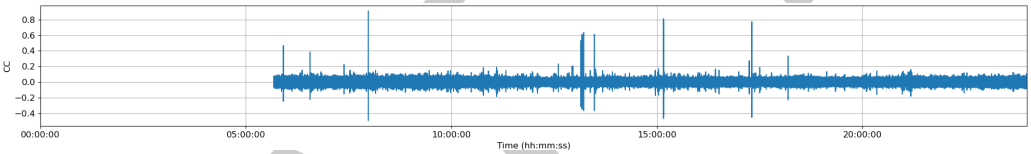


Figure 3: Cross correlation vector corresponding of the correlation between the extracted signal from the Figure 2, with the rest of the daily signal from the top panel of Figure 2

The RapidAlignerSeismic package contains tools for post-processing template matching results. These tools can clean up duplicate events, remove false detections, and cluster events. The script 'postProcess.py' constructs a list of unique detections from the raw detections obtained with RAS. Then, it computes the cross-correlation matrix of those events and generates a dendrogram to identify clusters of events based on their dissimilarity. A second script, 'display_cluster.py', uses the dendrogram to plot events grouped by cluster. Finally, it can generate a proto catalogue containing events date, cluster id, and event type (volcano-tectonic, tremors, teleseism, etc..), if this information is provided by the user.

This workflow can be used for the blind analysis of seismic data and the potential discovery of new types of signals.

Software design

Building on NVIDIA rapidAligner (Hundt et al., 2014), we have developed a Python package designed for processing long streams of three-component continuous seismic data (up to several years long). Our approach involves processing seismic data stored in a structured or unstrucured database in the miniseed format with minimum data preprocessing and user input. The I/O procedures use the Python obspy package (Beyreuther et al., 2010). When loaded with RAS, the code first applies a Hanning taper and a bandpass filter based on user inputs. Then it normalize the signal by its standard deviation. Gaps in the data are filled with random white noise with a standard deviation of MAD/1000. At this point, depending on user input, the data streams may be decimated or upsampled. Once loaded, the data are then organized according to the selected computing mode.

Depending on the selected mode, RAS will automatically detect the number of available GPUs on the computing node and parallelize the workload using MPI processes while performing appropriate load balancing. In Standard mode, RAS launches an MPI process for each available

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GPU. Parallelization is achieved by evenly distributing the templates across the different MPI processes. Each process then applies its subset of templates to the entire list of seismic data files. Since memory scales with the number of templates to be processed, user can set the number of templates to be processed per GPU at a time to prevent memory issues.

In Systematic mode, RAS launches two MPI processes for each GPU available. The idea here is that one MPI process maximizes the GPU occupancy while the other processes results: to avoid memory issues and reduce computing latency, once a day is processed, the first process sends the results to the second MPI process which is in charge to write the results to a dedicated file. In this mode, load balancing is ensured by distributing daily files accross the pairs of MPI processes according to the following pattern (considering a machine with two GPUs):

```
File 1 -> GPU 0
File 2 -> GPU 1
File 3 -> GPU 1
File 4 -> GPU 0
File 5 -> GPU 0
File 6 -> GPU 1
...
```

Research impact statement

RAS is currently being used in several current seismological research projects, covering a wide range of tectonic environments in different regions. For instance, RAS has been used to study the seismic activity beneath the Santiago Basin, providing new insights into the active faults that threaten this densely populated area (Ammirati et al., 2022). RAS is also being used to expand the catalog of recently discovered deep long-period (DLP) events beneath the Massif Central Volcanic Province. Thus far, we have increased the number of detected DLPs by more than an order of magnitude. These events are an important discovery because they suggest that volcanic activity is still present. RAS is also used to detect volcano-tectonic (VT) events beneath the Montagne Pelée in Martinique, which are important for monitoring volcanic activity.

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AI usage disclosure

No generative AI tools were used in the development of this software, the writing of this manuscript, or the preparation of supporting materials. A generative AI tool was used to enhance the wiki documentation and generate a first draft of functions docstring.

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